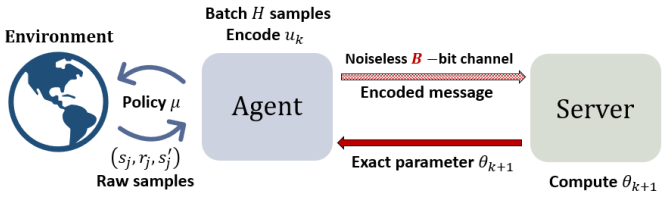
Adaptive Quantization Preserves Fast Rates for TD Learning over Rate-Limited Channels

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Problem and method. An agent evaluates a fixed policy μ in a finite discounted MDP using $\widehat{V}_\theta(s) = \phi(s)^\top \theta$. It sends TD estimates to a server through a noiseless B-bit uplink; the reverse channel is exact and unconstrained. **Batched Adaptively Quantized TD (BAQ-TD)** batches H samples at a frozen parameter and encodes innovations relative to the previous decoded direction. Batching reduces sampling noise; an adaptive encoding radius controls quantization error.



Assumptions and notation. The agent receives i.i.d. stationary tuples (s, r, s') under μ, with $s \sim \nu$ and $s' \sim P_\mu(s, \cdot)$. Assume $\|\phi(s)\| \le 1$, $\Sigma \succ 0$, $|R_\mu(s)| \le \bar{r}$, and conditionally σ^2-sub-Gaussian reward noise $r - R_\mu(s)$. Define the TD fixed point and conditioning parameter by

$$A := \mathbb{E}\Big[\phi(s)\big(\phi(s) - \gamma\phi(s')\big)^\top\Big], \quad b := \mathbb{E}[r\phi(s)], \quad \theta^\star := A^{-1}b, \quad \Sigma := \mathbb{E}[\phi(s)\phi(s)^\top], \quad \omega := (1-\gamma)\lambda_{\min}(\Sigma).$$

Write $T = KH$, $\eta := \sqrt{d}/2^m$, $\epsilon_H := C_\delta/\sqrt{H}$, and $C_\delta^2 := 2M\log(4Kd^2/\delta)$, where $M := \max\{4d^2, d(\bar{r}^2 + \sigma^2)\}$ and $\delta \in (0,1)$.

1. Batched estimation and adaptive innovation encoding. For epoch k, with $|I_k| = H$, compute

$$Y_k = \frac{1}{H}\sum_{j \in I_k} \phi(s_j)\big[r_j + \gamma\phi(s'_j)^\top\theta_k - \phi(s_j)^\top\theta_k\big], \qquad u_k = Y_k - h_{k-1}.$$

A scalar midpoint quantizer Q_{m,ρ_k} uses m bits per coordinate. The agent encodes $z_k = Q_{m,\rho_k}(\mathrm{sat}_{\rho_k}(u_k))$; the server updates

$$h_k = h_{k-1} + z_k, \qquad \theta_{k+1} = \theta_k + \alpha h_k, \qquad \rho_{k+1} = \eta\rho_k + 2\alpha\|h_k\| + 2\epsilon_H(\|\theta_k\| + 1).$$

Initialize $h_{-1} = 0$ and $\rho_0 = \bar{r} + 2\|\theta_0\| + \epsilon_H(\|\theta_0\| + 1)$.

2. High-probability statistical accuracy. Choose

$$\alpha = \frac{\omega}{32}, \qquad m = \left\lceil \frac{1}{2}\log_2\left(\frac{64d}{\omega}\right)\right\rceil, \qquad K = \left\lceil \frac{32c_1\log T}{\omega^2}\right\rceil, \qquad T = KH \ge \frac{c_2 K C_\delta^2}{\omega^2}.$$

For sufficiently large universal constants c_1, c_2, C, with probability at least $1-\delta$,

$$\|\theta_K - \theta^\star\|^2 \le \frac{V_0}{T} + \frac{C_p\log T}{T}\log\left(\frac{d^2\log T}{\omega^2\delta}\right), \qquad V_0 := \|\theta_0 - \theta^\star\|^2 + \alpha\rho_0^2, \quad C_p := \frac{Cc_1^2(1+\|\theta^\star\|)^2 M}{\omega^4}. \tag{1}$$

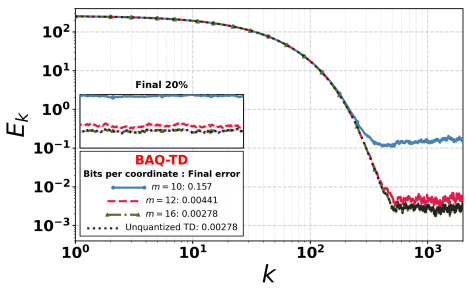
Thus BAQ-TD attains the $\widetilde{O}(1/T)$ squared parameter-error rate of unquantized TD, with a high-probability guarantee.

3. Communication cost. There is one $B = dm$-bit uplink message per epoch, giving

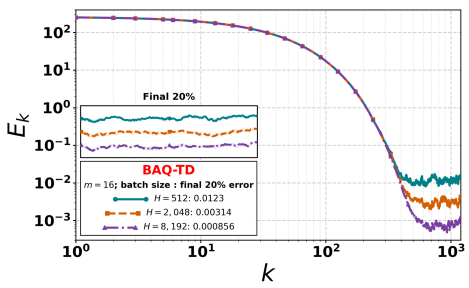
$$K = O\left(\frac{\log T}{\omega^2}\right) \text{ messages}, \qquad KB = O\left(\frac{d\log(d/\omega)}{\omega^2}\log T\right) \text{ total uplink bits.} \tag{2}$$

For fixed d, ω, this is $O(d\log T)$ uplink bits; the required bits per message are independent of T.

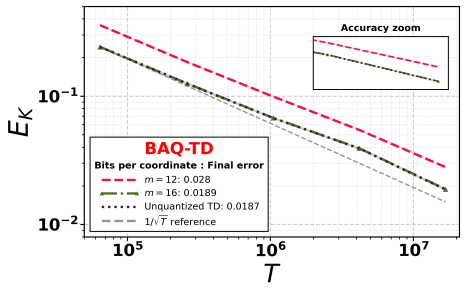
Numerical results. $|\mathcal{S}| = 100$, $d = 10$, $\gamma = 0.5$, $\bar{r} \le 5$, $\sigma^2 \le 1$; 100 independent runs. Plots report the mean error norm $E_k = \|\theta_k - \theta^\star\|_2$. In (a), $K = 1000$ and $m = 16$.



(a) Larger H lowers terminal error.



(b) $m = 16$ approaches unquantized TD.



(c) Observed $T^{-1/2}$ error scaling.